ENGN1218 Cheat Sheet

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1 Thevenin

Dependent sources only

- Plug in a VS/CS and find resistance.

Dependent and Independent Sources

- Open Circuit the Connections
- Deactivate Sources.
- Plug in a VS/CS and find resistance.

2 Conversion tricks

Y to Gamma

$$R_{\text{BC}} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C}$$

Gamma to Y

$$R_A = \frac{R_{\text{AB}} R_{\text{AC}}}{R_{\text{AB}} + R_{\text{AC}} + R_{\text{BC}}}$$

3 Capacitor/Inductors Equations

Natural Response/Step Response

The general form of all equations are

$$x(\infty) + (x(0^+) - x(\infty))e^{-\frac{t}{\tau}}$$

where x can be substituted with Voltage/Current. $\text{T} = R \times C$ or $\text{T} = \frac{L}{R}$

Capacitor Equations

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$$

$$I = C\frac{dV}{dT}$$

$$E = \frac{1}{2}CV^2$$

$$Q = CV$$

Inductor Equations

$$V = L\frac{dI}{dT}$$

$$i(t) = \frac{1}{L}\int_{t_0}^{t} i(\tau)d\tau + i(t_0)$$

$$E = \frac{1}{2}LI^2$$

$$I = QT$$

4 Rectifiers

$$V_{\text{RMS}} \times \sqrt{2} = V_p$$

$$V_{o(DC)} = V_{o(p)} - \frac{1}{2}V_{r(pp)}$$

$$V_{r(pp)} \simeq \frac{V_{o(DC)}}{2fR_LC} = \simeq \frac{V_{o(p)}}{2fR_LC}$$

For a Full-Wave Rectifier Circuit

$$V_{r(pp)} \simeq \frac{V_{r(pp)}}{fR_LC}$$

$$\gamma = \frac{V_{r(pp)}}{V_{o(DC)}} \times 100\%$$

- PIV means Peak Internal Voltage, $PIV = V_{in(P)} - 0.7$ or for a full wave rectifier $PIV = V_{in(P)} - 1.4$

5 Diode

Zener Diode

I_{ZK} Means Current Knee for a zener diode
I_{Zm} Means Current Maximum for a zener diode

6 Proofs/Questions

Ripple Factor

$$Q = CV$$
$$Q = CV_{r(pp)}$$
$$I = \frac{Q}{t}$$
$$Q = It$$
$$CV_{r(pp)} = \frac{I}{t}$$
$$V_{r(pp)} = \frac{It}{C}$$
$$V_{r(pp)} = \frac{It}{C}$$
$$I \simeq \frac{V_{o(p)}}{R_L}$$
$$t = \frac{1}{f}$$
$$V_{r(pp)} \simeq \frac{V_{o(p)}}{fR_LC}$$

For a full wave rectifier circuit, $t = \frac{1}{2f}$